Project Proposal: Optimizing Query Execution for Best-Selling Books Analysis: Bridging Logical and Physical Optimization, By: Nina Bryan (ninabry)

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Background and Motivation:

In this project, I aim to address two complementary tasks related to books and their metadata. The first focuses on improving the performance of complex SQL queries on large book datasets through query optimization techniques. The second involves leveraging computer vision to classify book genres based on cover images, using a pre-trained ImageNet model.

Efficient query optimization is a cornerstone of database management, especially when working with complex datasets such as best-selling books data. This dataset, which categorizes books by sales, genres, and authorship, offers an opportunity to explore advanced query execution techniques. The primary motivation for this project lies in ***reducing execution time*** for complex queries, such as identifying authors who have written books across diverse genres or finding those with similar characteristics to prolific writers like J.K. Rowling. Traditional query execution often suffers from inefficiencies like correlated subqueries and redundant scans, increasing computational costs. Query optimization operates at two levels: logical, which transforms queries into more efficient forms (e.g., reordering operations or replacing subqueries with joins), and physical, which selects optimal execution plans using techniques like indexing and caching. This project improves query performance by applying these strategies while demonstrating scalable and practical solutions for real-world data challenges.

Example of Physical Query Optimization Task:

```
SELECT * FROM R, S, T
 WHERE R.a = S.a
   AND S.b = T.b
```

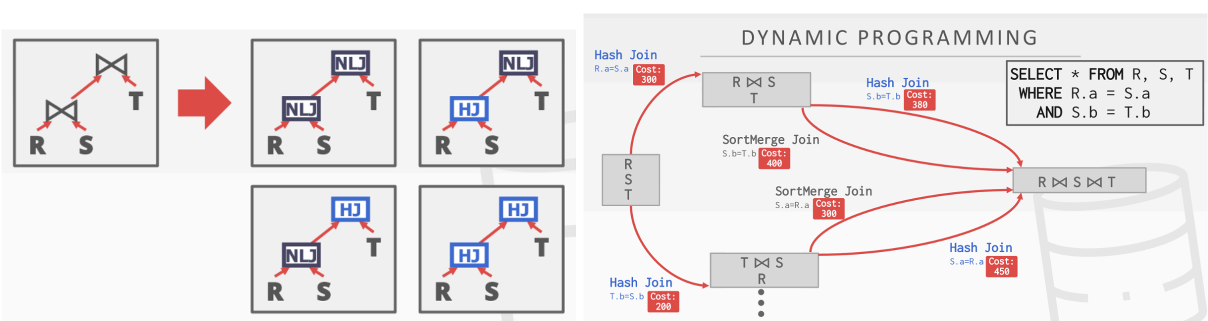

Figure 1. The diagram illustrates query optimization for a join query involving R, S, and T. Different join methods (e.g., Nested Loop, Hash Join) and orderings are evaluated, with dynamic programming used to compute and select the lowest-cost execution plan. *Textbook: Database Management Systems by Raghu Ramakrishnan and Johannes Gehrke*

Complementing the database-focused component, the second task of this project brings a visual and predictive dimension by analyzing book covers to classify genres. Book cover images are often the first interaction readers have with a book, and their design plays a crucial role in communicating genre and attracting audiences. Using the BookCover30 dataset, which contains 57,000 book covers across 30 genres, this part of the project employs deep learning techniques to predict genres based solely on visual cues. A pre-trained ImageNet model, such as ResNet or EfficientNet, serves as the backbone for feature extraction and classification, leveraging transfer learning to adapt to the book cover domain.

The combined tasks are highly relevant to the publishing and e-commerce industries, where efficient data processing and visually driven insights are essential. Query optimization ensures that metadata can be analyzed and retrieved swiftly, enabling services like search engines and recommendation systems to function seamlessly. Simultaneously, book cover classification can power applications such as personalized genre recommendations, dynamic content curation, and market trend analysis. Together, these components highlight how the integration of structured metadata and unstructured visual data can create robust, multifaceted solutions for modern challenges in data management and machine learning.

Literature Review:

- Study 1: Query Optimization

Query optimization in database management systems (DBMSs) involves selecting the most efficient access plan to process a query, as different plans can vary significantly in execution cost. The query optimizer comprises modules like the Rewriter, which transforms queries into more efficient forms, and the Planner, which uses strategies such as dynamic programming to explore access plans and select the least costly one. Supporting components include the Algebraic Space, which orders operators, the Method-Structure Space, which specifies execution methods, the Cost Model, which estimates plan costs, and the Size-Distribution Estimator, which predicts result sizes. While query optimization has been extensively studied, challenges remain, especially for advanced queries in areas like distributed and parallel systems, highlighting the field's ongoing relevance and complexity.

- Study 2: ImageNet: A Large-Scale Hierarchical Image Database

The ImageNet project introduced a large-scale, hierarchical image database designed to address challenges in organizing and utilizing the explosion of online image data. Built on the semantic structure of WordNet, ImageNet features millions of high-resolution, annotated images across thousands of categories, with plans to expand to 50 million images. The database emphasizes clean annotations, diverse content, and a densely populated hierarchy, making it a valuable resource for object recognition, classification, and clustering. Its construction relied on innovative methods like leveraging Amazon Mechanical Turk for labeling. ImageNet's scale, accuracy, and semantic richness set it apart from other datasets, offering unparalleled opportunities for advancements in computer vision and related fields.

- Study 3: [Are we done with Image Net?]

ImageNet's labeling approach has significant limitations, including assigning a single label to images with multiple objects, overly restrictive label proposals, and arbitrary class distinctions, which result in biased and noisy data. To address these issues, researchers proposed Reassessed Labels (ReaL) that better capture image diversity, multi-label training objectives, and a cleaned dataset to reduce noise and improve generalization. These enhancements, along with the new ReaL accuracy metric (instead of evaluating a model based on a single "correct" label per image, ReaL accuracy measures whether the model's top-1 prediction is included in a set of acceptable labels derived from diverse human annotations), provide a more meaningful evaluation of model performance, highlighting that recent models outperform original ImageNet labels and prompting the need for evolving benchmarks as models near-optimal performance.

Data Exploration and Preliminary Results:
The best-selling books dataset is relatively small (roughly 175 book titles), which simplifies processing but limits the statistical power of analyses. It also contains missing values in key attributes like Genre. Potential solutions include using imputation techniques (e.g., filling in genres based on similar books by the same author) or excluding incomplete records, depending on the impact of missing data on query goals. I may contemplate using this alternative [dataset].

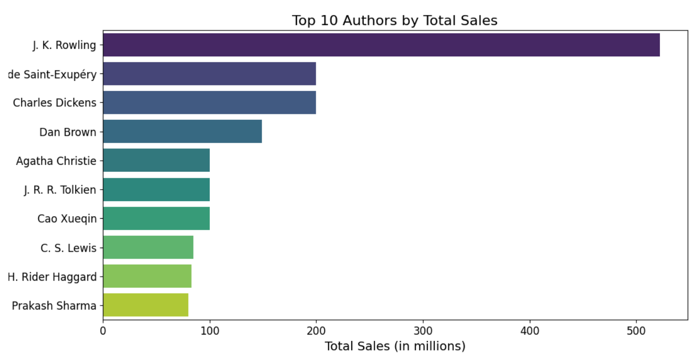

Figure 2. Top 10 Authors by Total Sales.
J.K. Rowling leads the list with significantly higher sales than the other authors in this dataset, likely due to the global success of the Harry Potter series. This insight could be valuable for forming queries, such as identifying authors who write in similar genres. Such data could help a publishing company strategize to replicate Rowling's success.

For the book cover dataset, the large size (~57,000 book covers) offers substantial training data for genre prediction but introduces challenges in data handling and storage since only image links are provided, not physical files. Given that the dataset was created in 2017, some links may be broken, requiring validation and downloading scripts to curate the dataset.

Project Methods:
For the best-selling books query task, tools like Python's sqlite3 module and time.perf_counter() will be used to analyze query performance. To further optimize query processing and enhance memory efficiency, I plan to implement a custom iterator class to handle batched result retrieval, a technique learned during this course. This approach will minimize memory usage and allow for scalable processing of query results, especially when dealing with large datasets or complex query outputs.

For the book cover analysis, tools like PyTorch for deep learning and OpenCV for preprocessing will be explored. GPU resources will be essential for model training due to the dataset's size, while bottlenecks may arise from downloading and preprocessing the images. Addressing these issues will ensure both datasets are ready for analysis and modeling.

Project Deliverables:

A successful project will produce optimized SQL queries with detailed analysis of their execution plans and measurable improvements in performance, as well as a trained and validated deep learning model for classifying book genres based on cover images. Sub-goals include implementing logical and physical query optimizations, such as creating indices and replacing correlated subqueries, and building a custom iterator class for efficient result processing. For the book cover task, sub-goals include curating and preprocessing the dataset, fine-tuning a pre-trained ImageNet model, and evaluating its performance with metrics like accuracy and F1-score. Together, these deliverables will demonstrate efficient data handling and predictive modeling.

Timeline:

Week 1-2:

- Analyze the best-selling books dataset for missing values and inconsistencies.
- Implement initial query optimizations, such as creating indices and replacing correlated subqueries with joins.
- Build and test a custom iterator class for efficient query result processing.
- Begin curating the book cover dataset by validating image links and downloading available files.
- Set up the environment for deep learning (install necessary libraries like PyTorch, OpenCV).

Week 3-4:

- Finalize query optimizations, including materialized views and late materialization techniques.
- Measure and document query performance improvements using tools like time.perf_counter().
- Preprocess the book cover images (resize, normalize) and create train-test splits.
- Fine-tune a pre-trained ImageNet model (e.g., ResNet or EfficientNet) on the BookCover30 dataset.
- Evaluate the model on the test set and identify bottlenecks or areas for improvement.

Week 5:

- Summarize and document query optimization techniques and results, including execution plan comparisons.
- Complete training of the genre classification model, focusing on hyperparameter tuning and visualization of results.
- Prepare the final project report including insights, challenges, and deliverables.